Virtual prototyping of the human brain: building a high resolution and scalable simulation that can aid the development of current and future electrical stimulating devices for the human brain.

Narrative

Our goal is to develop a modular computational framework for simulating the stimulation of human cortical neuronal responses that is designed to be extendable to a wide variety of current and future technologies. With the advent of the BRAIN initiative, as we move to new and finer scale methods of neuronal stimulation in humans, it will be critically important to develop computational frameworks of adequate scope and sophistication.

Abstract

With the advent of the BRAIN initiative, and as we move to new and finer scale methods of neuronal stimulation in humans, it will be critically important to develop computational frameworks of adequate scope and sophistication. This is not simply a scientific challenge, but a technical one, since these simulation frameworks will need to be extremely high resolution in both space and time.

As proof of concept, we will build stimulation modules designed to prototype electrical cortical stimulation of early cortical visual areas, since this technology is just beginning clinical trials. We will build modules to stimulate both surface electrodes (Orion, Second Sight Medical Products Inc.) and depth electrodes, since this is a technology likely to reach clinical trials in a few years. Our goal is to develop a modular simulation framework for human cortical stimulation that is designed to be extendable to a wide variety of current and future technologies. When applied to stimulation of visual cortex, the result will be a model that not only predicts neuronal responses at the population level, but also predictions of the induced visual percept.

Realistic simulation frameworks are a critical but understudied aspect of stimulation technology. Simulations can play a critical role in (1) Understanding the perceptual consequences of interactions between electronics and neurophysiology (2) Assessing effectiveness of current & future stimulation technologies, and (3) Finding better stimulation protocols for current technologies by providing a synthetic dataset for deep learning methods.

Virtual prototyping is now automatically used for almost any complex engineering design. It is time to extend these methods, so as to include both neural responses in the human brain and the resulting perceptual experience – creating 'virtual patients'. Traditional prototyping techniques can be then used to simulate a variety of stimulation technologies, thereby enabling evidence-based design decisions. Our group has already used this method successfully for retinal implant technology and are experts in cortical neuroimaging, so this proposal builds on considerable background expertise.

Summary

Damage to the optic nerve or lateral geniculate nucleus is responsible for 9/10 incidents of blindness. For these individuals, one promising approach for restoring vision is direct electrical stimulation of visual cortex, using either surface or penetrating electrodes. One visual cortical implant based on surface electrodes (Orion, Second Sight Medical Implants) has already received conditional FDA approval, with others scheduled to start clinical trials within a few years.

In the case of retinal prosthetics, it was quickly discovered, once these devices were implanted in patients, that there were substantial perceptual distortions due to interactions between the implant electronics and the underlying neurophysiology. These interactions are likely to be an even more important factor for cortical

implants, yet their influence has not been fully considered. A simple stimulation based on well established models of V1 cortical physiology is shown here.

Figure 1 shows predicted percepts elicited by two types of array: (1) a simulation approximating the Second Sight array, with electrodes of 260 μm diameter, separated by 520 μm and (2) an approximation of a higher resolution "Utah" penetrating array containing electrodes of ~30 μm at the tip, separated by 260 μm. In both cases we assume an 8x 8 array configuration. The top row shows polar angle assignment in (a) visual space co-ordinates (b) the folded right hemisphere and (c) the flattened cortical surface. The approximate location of the simulated arrays, near the foveal pole, are shown with an asterisk in each representation. The middle row represents ~20mm of flattened cortex corresponding to the central 1.25° of visual angle on which we simulate (a) orientation 'pinwheels', (b) ocular dominance columns and (c) electric fields on the cortical surface for the outer 4 electrodes for "Utah" (tiny yellow, separated by 2.8 mm) and "Second Sight" (white, separated by 3.64 mm) arrays.

The third and fourth rows show simulated predicted phosphenes in visual space co-ordinates. Cortical prosthesis development is currently implicitly driven by (g, third row) a 'linear scoreboard' model in which it is assumed that the size of the apparent percept is is determined by the projection of the electrical field stimulation on the cortical surface into visual space co-ordinates. According to this model, small electrodes produce small discrete percepts. Incorporating (h, bottom row) simple well-established facts about orientation tuning, ocular dominance and receptive field sizes in V1 into the simulation produces dramatically different predictions – the phosphenes elicited by each of the 4 electrodes are hugely expanded and individual electrodes can no longer be resolved. It is immediately apparent that one of the fundamental assumptions of the scoreboard model – that reducing electrode size will increase perceptual resolution – is utterly wrong. Even in the fovea, the majority of neurons have receptive fields are ~1° of visual angle. Without a way of selectively stimulating neurons with small receptive fields or particular phase preferences, neuronal receptive field size rather than electrode size will limit perceptual resolution unless electrodes are separated by many millimeters of cortex. Comparing the Utah and Second Sight arrays in (h), it is also immediately apparent that smaller electrodes will selectively stimulate orientation and (depending on electrode depth) ocular dominance columns. As a result, small electrode sizes are likely to result in highly confusing perceptual experiences, analogous to presenting binocularly rivalrous stimuli (different stimuli in each eye).

Aim 1: To develop a realistic simulation of cortical prostheses

We will combine human patient psychophysics and computational modeling to develop a realistic model of the predicted effects of electrical stimulation in V1-V3.This will include

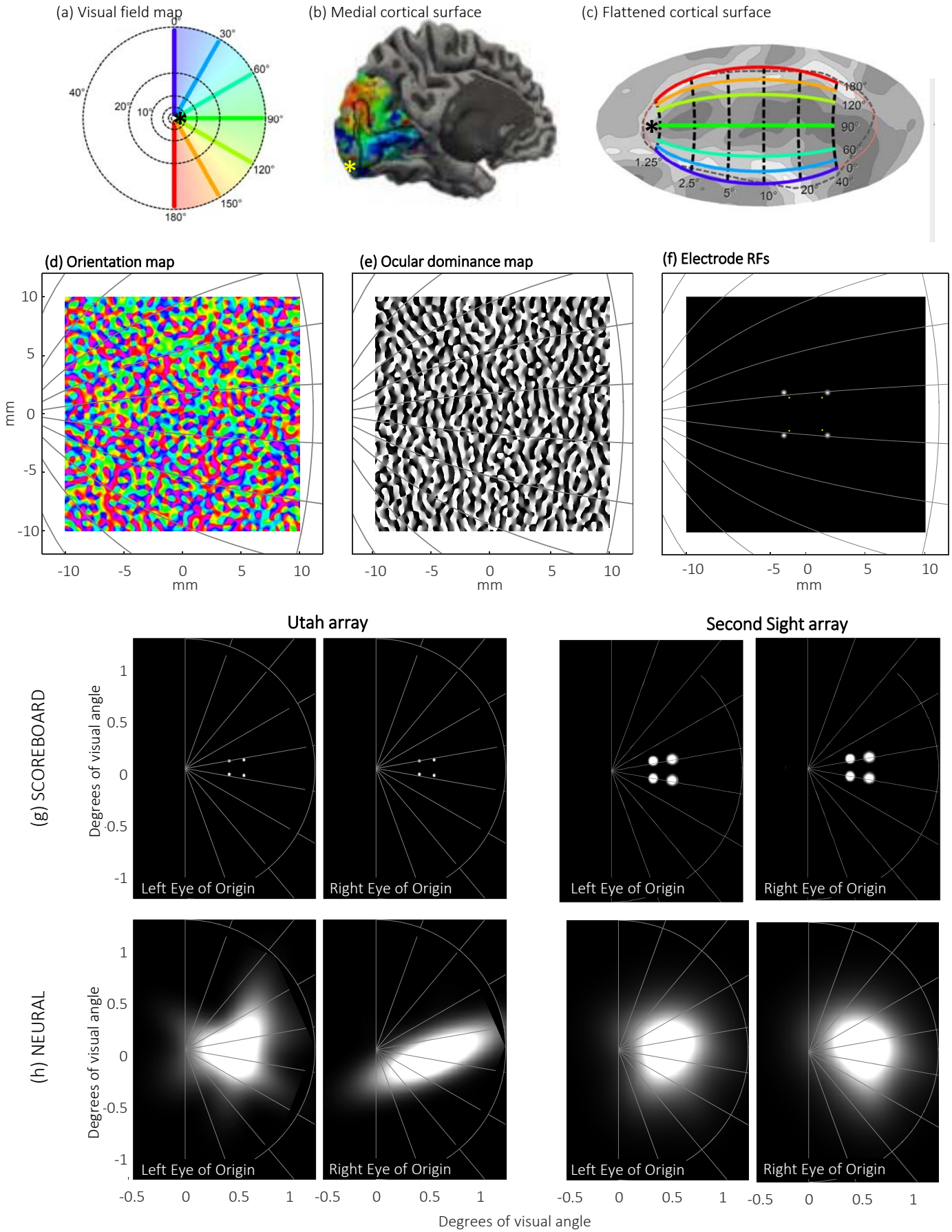
- Implementing a finite element model of electric fields for both surface and depth electrodes
- Implementing a transform between individualized cortical surface anatomies as measured using standard MR imaging (T1 anatomies) and the retinotopic organization of areas V1-V3 (based on the Benson template). This will provide a useful guide to surgeons, allowing them to optimize array placement in patients based on individual early visual cortex anatomy.
- Developing a neuronal population model that describes the responses of early visual cortex neurons in V1-V3 to electrical stimulation.
- Developing Bayesian decoding model that finds the expected perceptual experience of the patient given the population of cell responses.

This forward model can be used to predict both neuronal responses and the perceptual experience of current and future electrical stimulation technologies.

Aim 2. Improving stimulation protocols

Using synthetic training sets generated by the model, we will use a deep neural network to learn an approximate inverse transformation that finds the best possible stimulation protocol to elicit a given desired percept.

We will compare these outcomes across different array configurations in order to find technologies that are likely to maximize perceptual outcomes.


(a) Visual field map
0°
30°
60°
90°
120°
150°
180°
40°
20°
10°
(b) Medial cortical surface
(c) Flattened cortical surface
180°
120°
90°
60°
0°
40°
20°
10°
5°
2.5°
1.25°
(d) Orientation map
(e) Ocular dominance map
(f) Electrode RFs
mm
Utah array
Second Sight array
(g) SCOREBOARD
(h) NEURAL
Degrees of visual angle
Left Eye of Origin
Right Eye of Origin